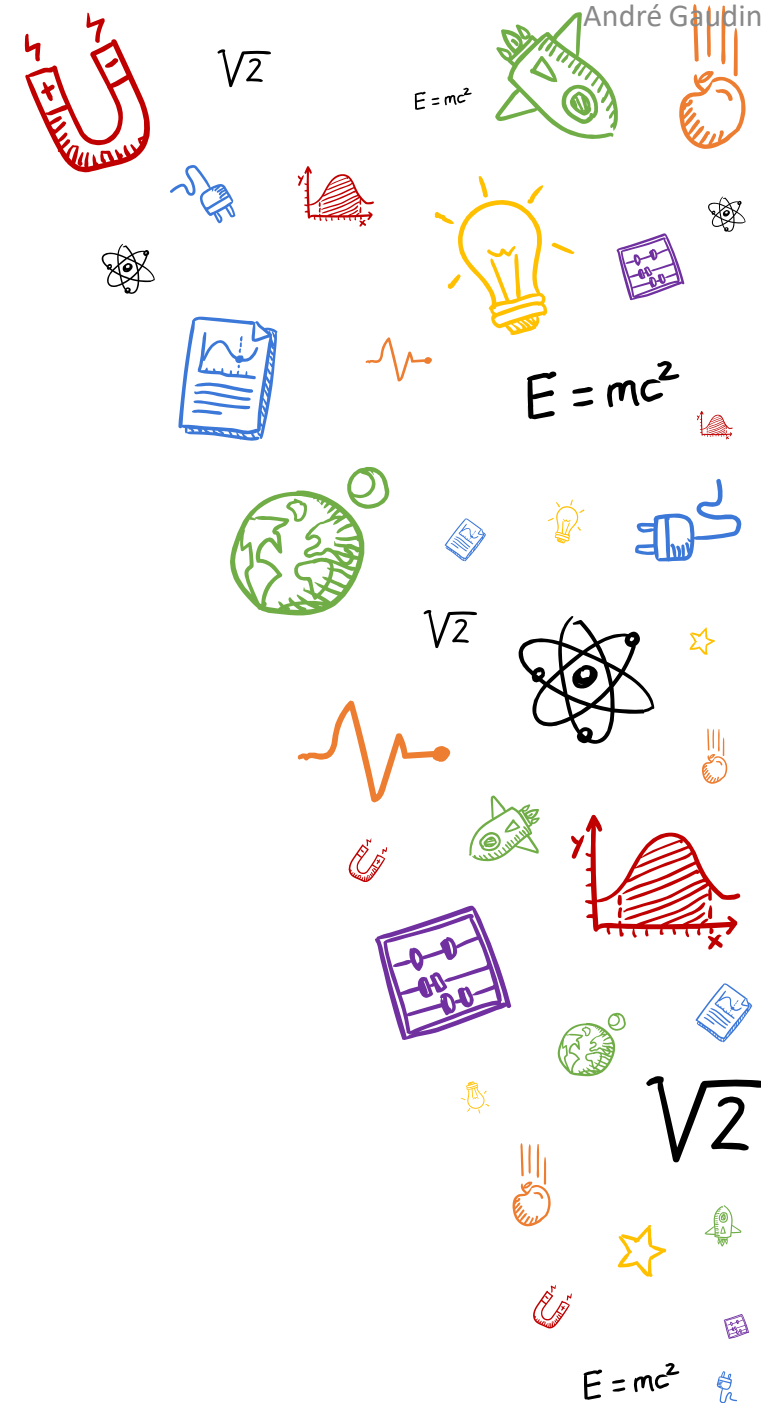


Forces and Newton's Laws of Motion

November 1st, 2023



This Week and Today



This Week

- Newton's Laws of Motion
 - ~~First Law~~
 - ~~Second Law~~
 - Third Law
- Surface Friction and Viscous Drag
- Elastic Restoring Forces and Hooke's Law
- Free Body Diagrams

Today

1. Surface Friction and Viscous Drag
2. Elastic Restoring Force and Hooke's Law
3. Newton's Third Law
4. Free Body Diagrams

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Surface/Solid Friction



You will **explore the force of friction** between some **wooden carts** (blocks) and **various surfaces**.



This may look simple, but you will have to approach it carefully to see each effect clearly and all the details of how the force changes.

Materials:

- 1 LabQuest 2 & 1 force sensor
- 1 wooden “cart”
- 1 each of: 1 kg and 500 g masses
- 1 tabletop (yes, that’s a surface too!)
- 1 sheet of mylar
- 1 sheet of plexiglass or 1 corkboard



The **mylar sheets** will be used later by the **Idea Lab** to make things, **please be careful** with them and **do not damage them** in any way.

Surface/Solid Friction



Goals:

1. What is the general shape of curve?
2. How does the nature of the surface affect the force of friction?
3. How does the mass of the “cart” affect the force of friction?

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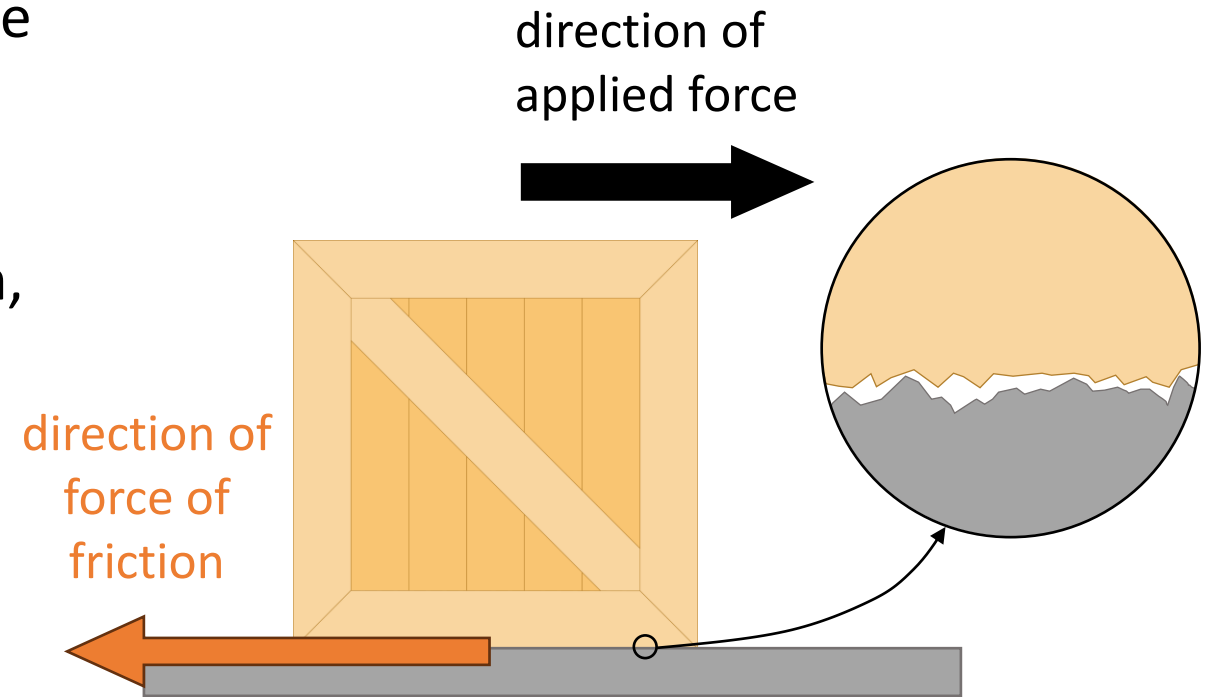
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Solid/Surface Friction



Caused by:

- **Peaks and valleys** at the microscopic/atomic scale. Force is needed to break/slide over/etc.
- Much of is due to **adhesive force** between the surface molecules
- Smooth surfaces, especially perfectly smooth, will “**cold weld**”



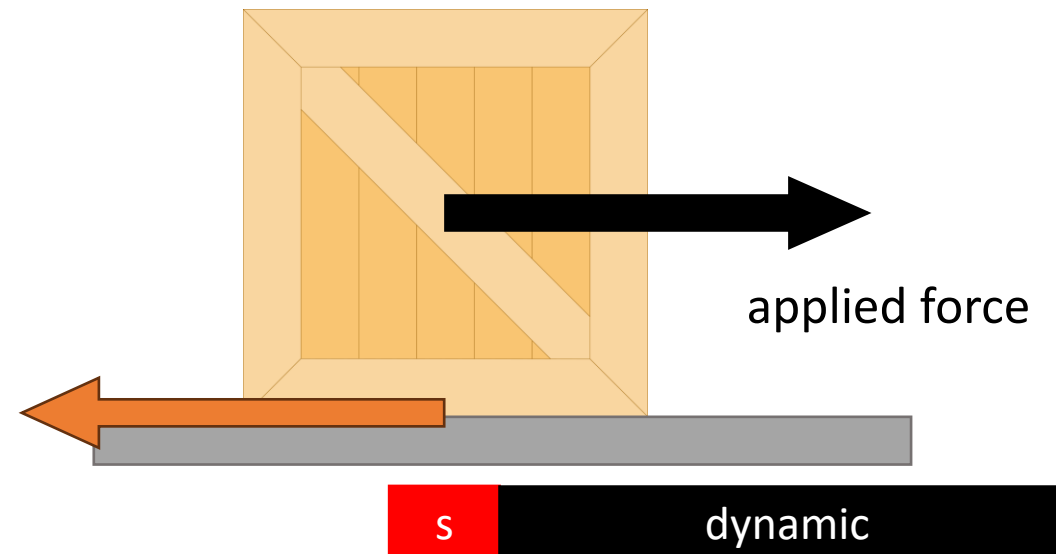
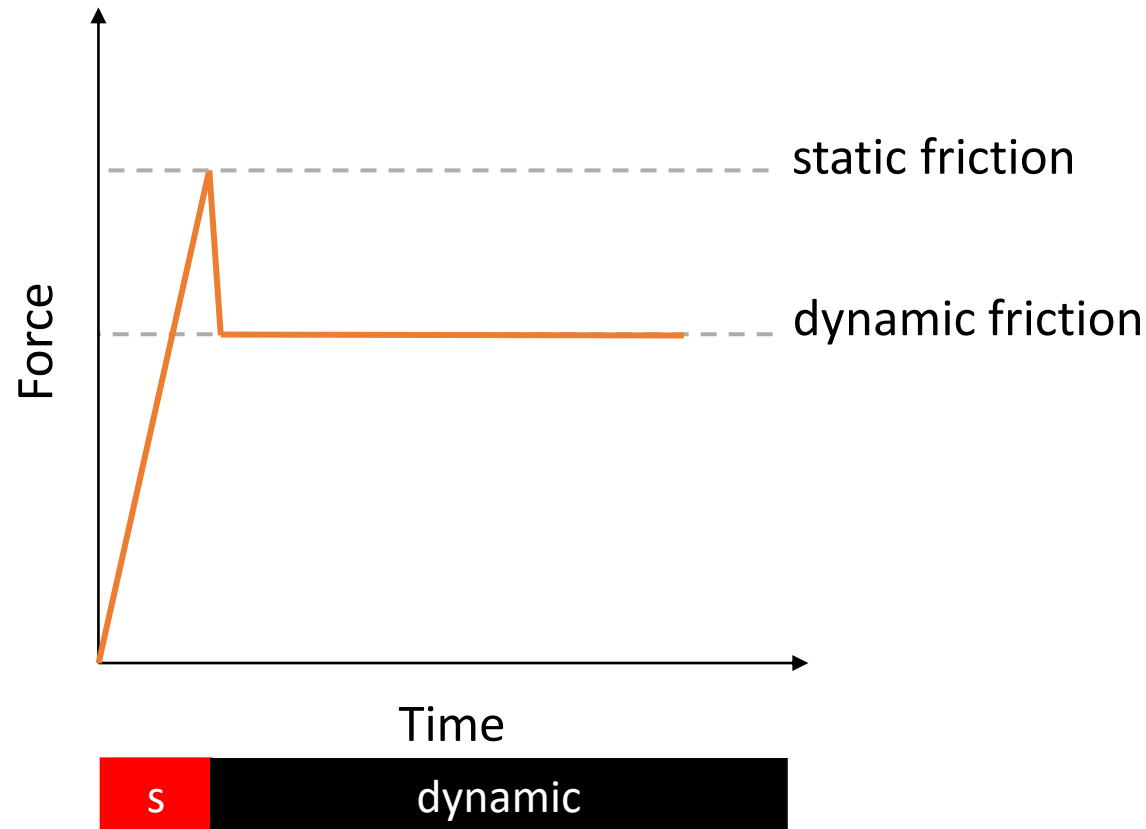
Solid/Surface Friction



When we try to move an object along a surface:

static friction: no motion, F_s

dynamic friction: motion, F_d



Solid/Surface Friction

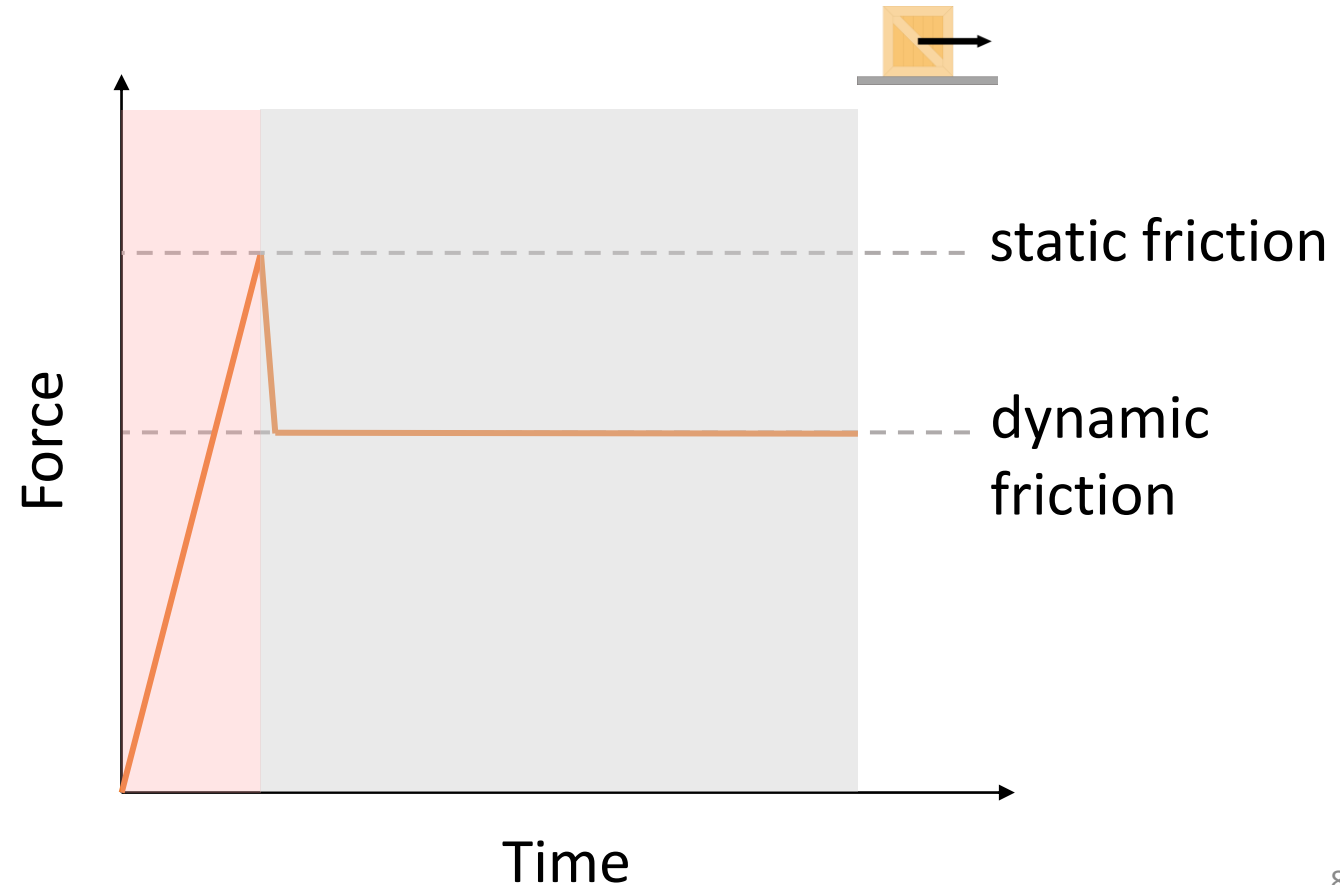


- F_s exactly matches the force applied
- Increases to a **maximum** $F_{s,max}$
- Friction **drops abruptly** to F_d
- Relationships:

$$0 \leq F_s \leq F_{s,max}$$

$$F_d \leq F_{s,max}$$

$$F_d = \text{const.}$$



Coefficients of Friction



- Coefficients of friction capture the degree of roughness or smoothness and the bonding and are the
- Two **types of friction** = **two coefficients** of friction

coefficient of static friction: μ_s

coefficient of dynamic friction: μ_d

- **Depends on the normal force, F_N**

Static:

$$F_f \leq \mu_s F_N$$

Dynamic:

$$F_f = \mu_d F_N$$

Coefficients are
unitless ratios:

$$\mu = \frac{F_f}{F_N}$$

Coefficients of Friction



Surface 1	Surface 1	μ_s	μ_d
glass	metal	0.7	0.6
rubber	concrete	1.0	0.8
rubber	wet tarmac	0.6	0.4
rubber	Ice	0.3	0.2
metal	metal (lubricated)	0.15	0.15

Static:

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Dynamic:

$$F_f = \mu_d F_N$$

Coefficients are
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Solid/Surface Friction



To stop your car of mass 1500 kg, you step on the brakes stopping the wheels from turning.

What is the force of friction?

Note: this assumes that the wheels lock up, normally stopping occurs because of the friction in the breaks and the wheels keep spinning.

Calculate the normal force:

$$F_N = mg = 1500 \text{ kg} \times 9.8 \text{ ms}^{-2} = 14\,700 \text{ N}$$

Now we can calculate the force of friction:

$$F_f = \mu_d F_N = 0.8 \times 14\,700 \text{ N} = 11\,760 \text{ N} \approx 12\,000 \text{ N}$$

Surface 1	Surface 1	μ_s	μ_d
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Solid/Surface Friction



A piece of wood with a coin on it is raised on one end until the coin just begins to slip. The angle the wood makes with the horizontal is $\theta = 15^\circ$.



What is the coefficient of static friction?

Just at the point before moving, we know that the sum of the forces in both x and y direction is 0 N:

$$\begin{aligned} F_N - F_{gy} &= 0 \\ F_N &= F_{gy} \end{aligned}$$

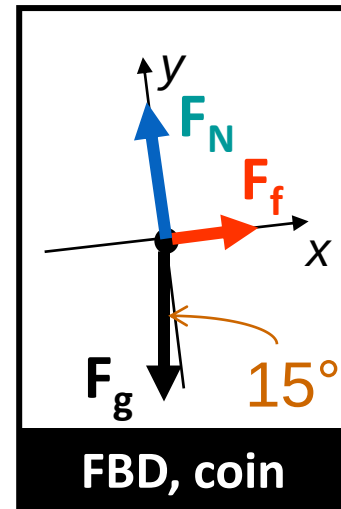
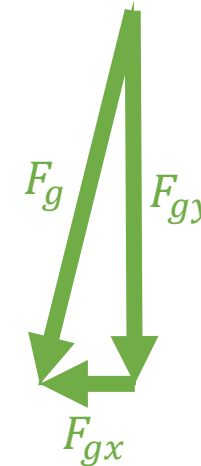
$$\begin{aligned} F_f - F_{gx} &= 0 \\ F_f &= F_{gx} \end{aligned}$$

Solving for the normal and force of friction:

$$\begin{aligned} F_N &= F_{gy} \\ &= mg \cos 15^\circ \\ F_f &= F_{gx} \\ &= mg \sin 15^\circ \end{aligned}$$

So the coefficient of friction is:

$$F_f = \mu_d F_N \quad \mu_d = \frac{F_f}{F_N} = \frac{mg \sin 15^\circ}{mg \cos 15^\circ} \approx 0.213$$



Translational Equilibrium



For an object being slide along a surface, what does translational equilibrium look like?

$$F_s = F_{\text{applied}} \text{ or } F_d = F_{\text{applied}}$$

The object is either not moving or sliding along at a constant velocity.

$a = 0 \rightarrow \sum F = 0$ is the **condition for translational equilibrium.**



Floating and Sinking



Why does an object sink in water etc.?

Buoyancy Force

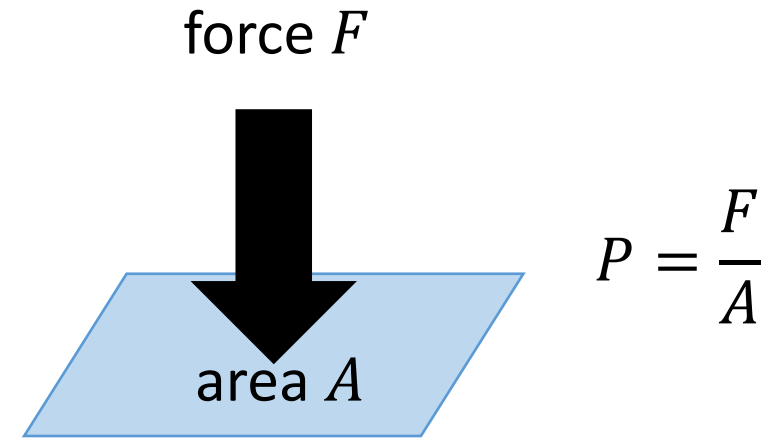


The **buoyancy force** is the force exerted by a fluid or gas on an object in or partially inside of it.

Density and pressure are important for buoyancy:

- density is in kg/m^3
- pressure the force per unit area $P = F/A$
and units (pascals) are $\text{N/m}^2 = \text{Pa}$

The buoyancy force comes from the **difference in pressure acting on the top and bottom** of an object.



Buoyancy Force



Three potential situations:

- **neutral buoyancy:** equal densities
- **sinking:** when **(overall average) density is greater** than fluid (net force down)
- **float:** **(overall average) density is less** than fluid, volume displaced is volume below the fluid's surface

The equation for the buoyancy force is:

$$F_b = \rho V g$$



where:

ρ – is the Greek letter rho, and the density of the fluid

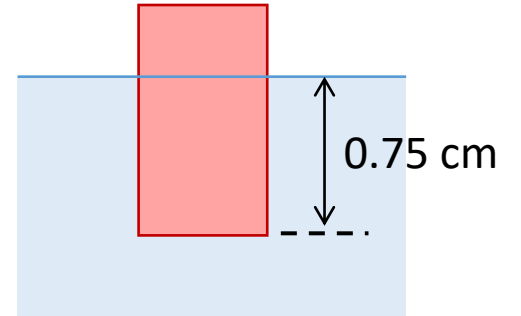
V – is the volume of fluid displaced by the object

g – is the acceleration due to gravity

Buoyancy Force



A buoy of a uniform cross-sectional area of 0.20 m^2 floats in fresh water, with 0.75 m of the height submerged below the water line. The density of fresh water is 1000 kgm^{-3} .



a. Calculate the mass of the buoy.

We know the object isn't moving up or down so:

$$F_b = mg$$

Then:

$$F_b = \rho V g$$
$$mg = \rho V g$$

$$m = \rho V = 1000 \text{ kg m}^{-3} \times 0.20 \text{ m}^2 \times 0.75 \text{ m} = 150 \text{ kg}$$

b. The buoy is now placed in seawater of a greater density than fresh water. Explain the change, if any, in the submerged height of the buoy.

The buoy will float higher because the higher density of water means less needs to be displaced to equal the mass of the buoy.

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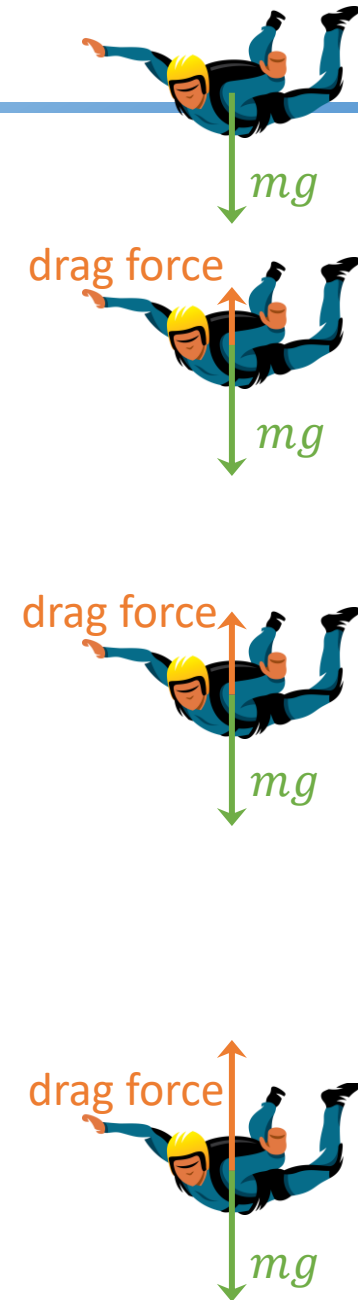
Air Resistance



Consider a skydiver:

- **Starts off** with **near zero speed** and so near zero **drag force** $F_d \approx 0$
- **Slowly builds larger** and larger **countering** the force of gravity as the forces are in opposite directions
- Eventually **net force** $F_g + F_d = 0$ and stops accelerating
- **Terminal speed** is the speed at this point

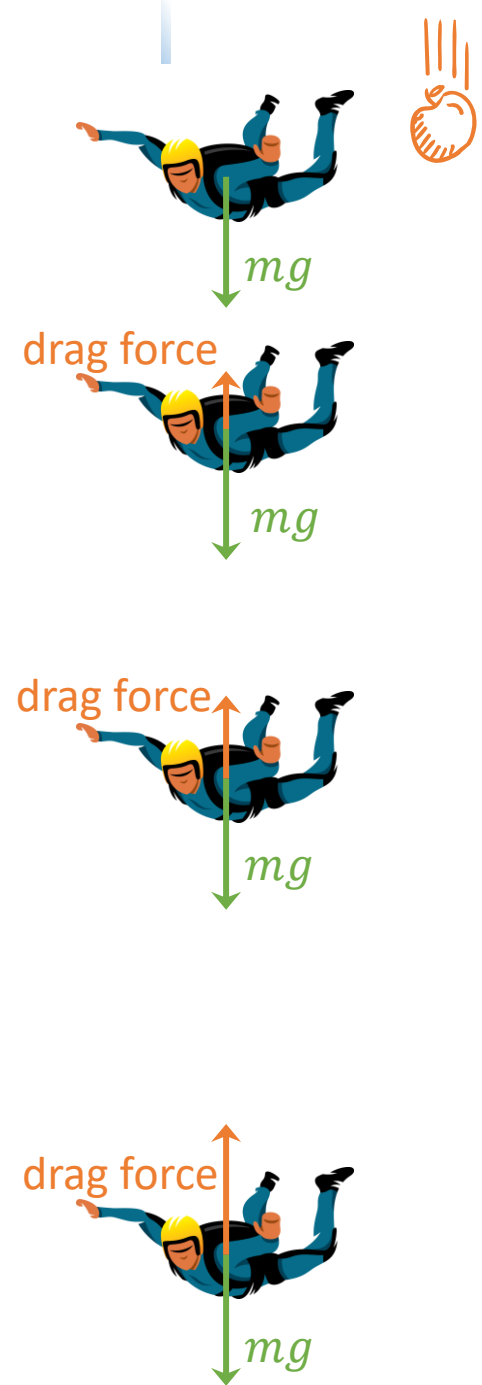
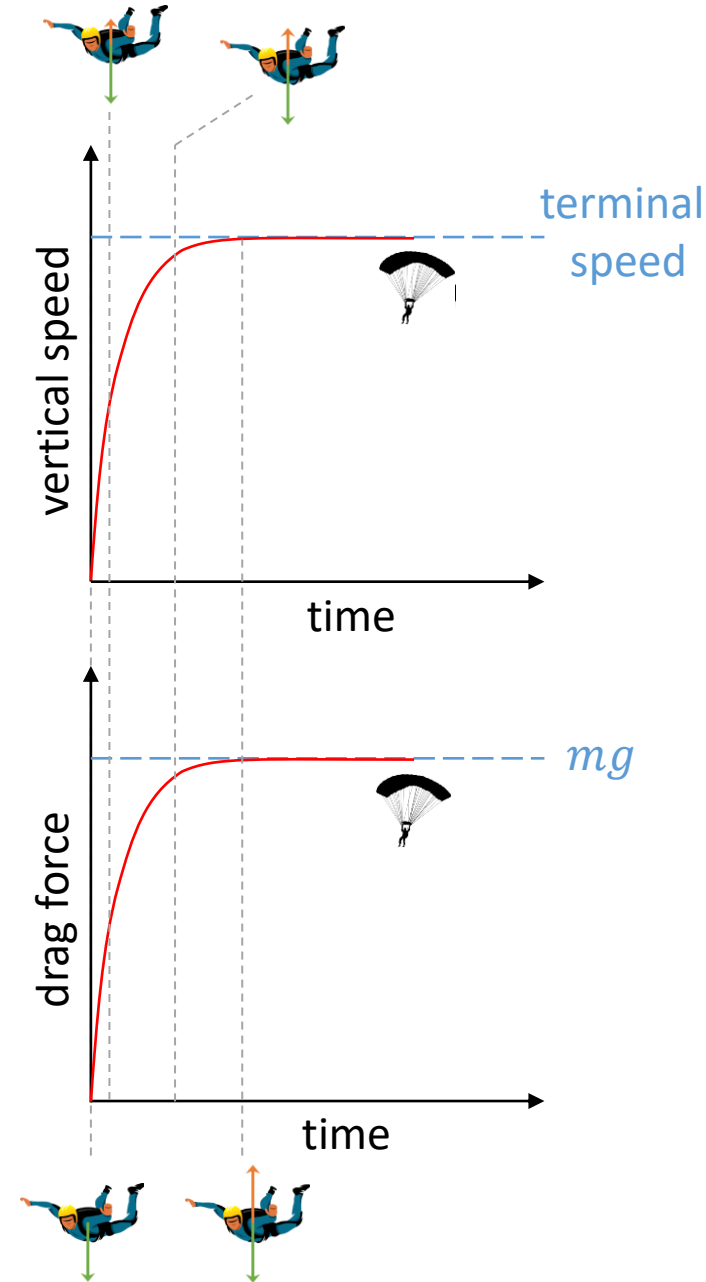
Terminal speed is the maximum a object can fall at in a given fluid or gas.



Air Resistance

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Stokes' Law

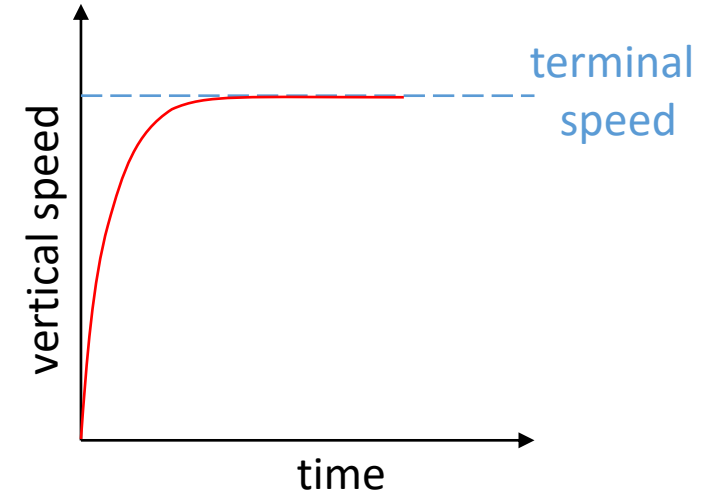


We can use Stokes' Law (George Stokes 1851) to estimate the terminal speed of a falling object.

- **Stokes' Law:**

$$F_d = 6\pi\eta r v$$

calculates the drag force (viscous force). Here **η** is **the dynamic viscosity** of the fluid with units of Pa s (poise)



Thinking of forces, what condition can we write when an object is moving at terminal speed?

Stokes' Law and Terminal Speed



Thinking of forces, what condition can we write when an object is moving at terminal speed?

We also have:

$$F_d = 6\pi\eta r v$$

At terminal speed, its speed is constant, so the net force is zero:

$$F_g - F_d - F_b = 0$$

$$F_g = F_d + F_b$$

Then:

$$mg = 6\pi\eta r v + \rho_f V g$$

$$\rho_o V g = 6\pi\eta r v + \rho_f V g$$

$$6\pi\eta r v = \rho_o V g - \rho_f V g$$

$$6\pi\eta r v = \rho_o V g - \rho_f V g$$

$$v = \frac{(\rho_o - \rho_f)gV}{6\pi\eta r}$$

$$v_t = \frac{(\rho_o - \rho_f)gV}{6\pi\eta r}$$

Stokes' Law and Terminal Speed



A basketball of radius $r = 11 \text{ cm}$ and density $\rho_s = 89.6 \text{ kgm}^{-3}$ is falling through the air.

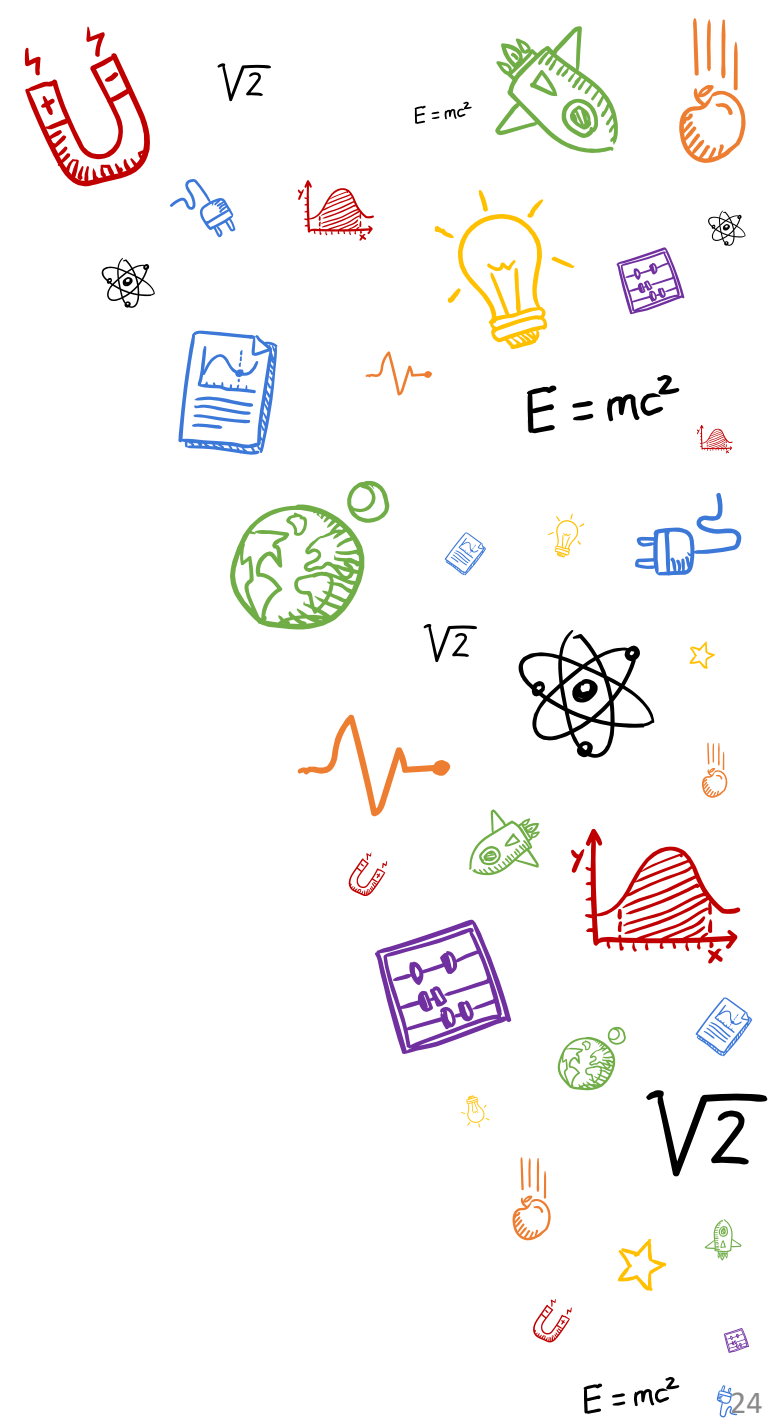
If the density of air is $\rho_f = 1.2 \text{ kgm}^{-3}$ and the viscosity is $\eta = 0.01 \text{ Pa s}$, what is the terminal speed of the basketball?



$$V = \frac{4}{3}\pi r^3 \approx 0.00557 \text{ m}^3$$

$$\begin{aligned} v_t &= \frac{(\rho_o - \rho_f)gV}{6\pi\eta r} = \frac{(89.6 \text{ kgm}^{-3} - 1.2 \text{ kgm}^{-3}) \times 9.8 \text{ ms}^{-2} \times 0.00557 \text{ m}^3}{6\pi \times 0.01 \text{ Pa s} \times 0.11 \text{ m}} \\ &= 232.9438 \text{ ms}^{-1} \\ &\approx 230 \text{ ms}^{-1} \end{aligned}$$

Elastic Restoring Forces



Elastic Restoring Force and Hooke's Law



When you compress or stretch out a solid object by applying a force, for example a spring.

- Called the **elastic restoring force** F_H
- **Equal and opposite** to the force applied
- Proportional to the **extension**, x

$$F_H = -kx$$



- Called **Hooke's law**, where k is the **spring constant** (elastic constant, or force constant)

